

4.6.1 Elastic collisions

Studies of the collisions among objects, first made in the 17th-century, led to the discovery of two basic laws of Nature: the conservation of linear momentum, and the conservation of kinetic energy associated with a special class of collisions called *elastic collisions*.

The conservation of linear momentum in an isolated system forms the basis for a quantitative discussion of all problems that involve the interactions between particles. The present discussion will be limited to an analysis of the elastic collision between two particles. A typical two-body collision, in which an object of mass m_1 and momentum $\mathbf{p}_1$ makes a grazing collision with another object of mass m_2 and momentum $\mathbf{p}_2$ ($p_2 < p_1$), is shown in the following diagram. (The coordinates are chosen so that the vectors $\mathbf{p}_1$ and $\mathbf{p}_2$ have the same directions). After the collision, the two objects move in directions characterized by the angles θ and ϕ with momenta $\mathbf{p}_1'$ and $\mathbf{p}_2'$.

